American Computer Science League

2022-2023 ● Contest 2: Short Problems ● Intermediate Division

1. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if all numbers are single digits:

6 3 / 2 2 ^ + 3 4 2 – * /

A. 0
B. 1
C. 6
D. 12
E. 15

2. Prefix/Infix/Postfix Notation

Translate the following postfix expression into prefix if all numbers are single digits:

4 6 + 7 5 – / 2 ^ 7 5 4 – * +

A. +^/4+6–752*7–54
B. +^/+46–752*75–4
C. +^+/46–752*7–54
D. +^/+467–52*7–54
E. +^/+46–752*7–54

3. Bit-String Flicking

Evaluate the following expression:

(LSHIFT-2 (RCIRC-1 (NOT 10110)))

A. 01001
B. 10000
C. 00101
D. 10010
E. 01100

4. Bit-String Flicking

Solve the following expression for X if X is a 5-bit string:

(NOT 11011) AND X OR (RCIRC-1 01110) = 00111

A. 001*0
B. *0111
C. 00*00
D. *****
E. 01010

5. LISP

Evaluate the following LISP expression:

(CDR (CAR (CDR '(b (c (a) b)))))

A. (a b)
B. (c (a))
C. ((a) b)
D. (a)
E. (c (a) b)

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1. 前缀/中缀/后缀

如果下方所有数字都是一个单独的数值，请计算下方后缀表达式：

6 3 / 2 2 ^ + 3 4 2 – * /

A. 0
B. 1
C. 6
D. 12
E. 15

2. 前缀/中缀/后缀

如果下方所有数字都是一个单独的数值，请将下方**后**缀表达式转换为**前**缀表达式：

4 6 + 7 5 – / 2 ^ 7 5 4 – * +

A. +^/4+6–752*7–54
B. +^/+46–752*75–4
C. +^+/46–752*7–54
D. +^/+467–52*7–54
E. +^/+46–752*7–54

3. 比特位操作

请**计算下方**表达式：

(LSHIFT-2 (RCIRC-1 (NOT 10110)))

A. 01001
B. 10000
C. 00101
D. 10010
E. 01100

4. 比特位操作

X 是一个 5 位的位串，请根据下方表达式计算 X 的值。

(NOT 11011) AND X OR (RCIRC-1 01110) = 00111

A. 001*0
B. *0111
C. 00*00
D. *****
E. 01010

5. LISP 语言

请**计算下方** LISP 表达式：

(CDR (CAR (CDR '(b (c (a) b)))))

A. (a b)
B. (c (a))
C. ((a) b)
D. (a)
E. (c (a) b)